Ricardo (Zhicong) Xie

GRADUATE MECHANICAL ENGINEER | DESIGN, PROTOTYPING AND TEST | BUILDING SERVICES
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PROFILE

- Mechanical engineering master's student who designs, builds and tests real hardware. I designed and built the suspended-load platform of a wearable energy-harvesting backpack, wired its generator, traced a wiring fault and ran the power test that measured **55 mW against a 30-50 mW target**. Building services coursework (MECHENG 715), an NZ ventilation design and CFD/FEA experience.
- **Work rights:** completing my MEngSt in July 2027 and planning to apply for a Post Study Work Visa after graduation. Work rights are subject to visa approval.
- **Availability:** mechanical project internship in Shanghai, Nov 2026 - Feb 2027 (email preferred; video interviews welcome); back in Auckland for Semester 1 2027; available for graduate roles from mid-2027 and 2028 graduate programmes.

EDUCATION

University of Auckland | Master of Engineering Studies (Mechanical) Feb 2026 - Jul 2027 (expected)
GPA 6.0/9.0 | Relevant coursework: MECHENG 715 Building Services (Semester 2, 2026), Advanced Computational Fluid Dynamics (A–), Research Methods (A), Composite Materials

Huazhong Agricultural University, China | Bachelor of Engineering (Mechatronic Engineering) 2021 - 2025
Mechanical design, electrical and electronic engineering, sensors, embedded control, mechatronic integration

EXPERIENCE

Upcoming: Mechanical Project Mentor (Internship) | Mfeducation, Shanghai Nov 2026 - Feb 2027

Energy-Harvesting Backpack Prototype | Lihua Tang research group, University of Auckland Jul - Oct 2026

- **Design:** designed the suspended-load platform in SolidWorks: twin guide rods, parallel elastic bands and a rack-and-pinion drive that converts load movement into generator rotation. Kept a fixed mounting-hole pattern so the drive can be upgraded without recutting the plates.
- **Build:** laser-cut the acrylic plates, 3D-printed parts, welded the generator support structure and assembled the platform; found a printed-hole/M5 clearance problem, corrected the tolerance and updated the CAD.
- **Wiring and fault-finding:** wired every coil of the research group's generator prototype to a single output. When it gave no output, checked continuity joint by joint with a multimeter and traced the fault to a **coil lead worn through by friction** on an internal contact surface; repaired it and restored output.
- **Power testing (my own, 21 Oct 2026):** 3.3 kΩ load and oscilloscope capture. Walking at 4 km/h with the backpack on, it delivered **55 mW average, above the 30-50 mW project target** (81 mW hand-cranked; about 205 mW over the first 39 s at 5 km/h). Calculated average load power point by point from raw samples. Earlier took part in shaker-table tests and phone-charging and pulse-oximeter demonstrations led by a doctoral researcher.

PROJECTS

NZ Home Ventilation with Heat Recovery (MVHR) | Self-directed building services project 2026

- Designed whole-house ventilation for a New Zealand home: airflow schedule (60 L/s supply, 45 L/s extract, 85 L/s boost check), duct velocity and pressure-loss checks, preliminary equipment selection, a design-review issue log and a commissioning test plan, with notes against NZBC G4 and H1.
- Built the Revit MEP model with a **C# Revit API script** that generates the duct layout, airflow schedule and drawing sheets; exported plan, 3D and schedule sheets for review.

Solar Chimney Natural Convection and Conjugate Heat Transfer | ANSYS Fluent / CFX Course research project

- Built a 2D axisymmetric three-region conjugate heat transfer model of buoyancy-driven flow; mesh independence across three grids (36,847-cell production mesh, 0.90% outlet-velocity difference to the fine mesh).
- Found that the reference paper's 12.2 m/s draught would need a ~299 K temperature rise, inconsistent with its reported 42-46 °C, and treated it as an uncertain benchmark in the technical report.

Composite Front Wing FEA | Abaqus, Formula SAE-style load cases Course project

- S4R shell and composite layup model under aerodynamic, lateral and 4.1 g bump cases; identified the governing case with ~74% peak stress utilisation.

Automated Dynamic Weighing of Pig Carcasses | STM32, HX711 Sep 2022 - Jan 2024

- Developed load-cell acquisition, embedded logic and serial/Bluetooth transfer for weighing pig carcasses in motion on a processing line; the project software is registered as Software Copyright No. 2024SR0141888 (holder: Huazhong Agricultural University).

SKILLS

- **Mechanical design:** SolidWorks (proficient: parts, assemblies, drawings); AutoCAD/DXF (working)
- **Building services:** airflow schedules, duct sizing and pressure-loss checks, equipment selection, commissioning plans; NZBC G4/H1; Revit MEP model automation in C# (Revit API)
- **Simulation:** ANSYS Fluent/CFX and Abaqus (working); MATLAB/Simulink (basic)
- **Prototyping and test:** oscilloscope power measurement, multimeter fault-finding, arc welding and electronics soldering, laser cutting, 3D printing, CNC training, mechanical assembly
- **Programming:** Python (real-time device-control IoT scripts in a UoA postgraduate course, data processing); C# (Revit API); embedded C on STM32 (basic)
- **Languages and licence:** English (IELTS 7.0), Mandarin (native); full driver licence (China)

LEADERSHIP AND COMMUNITY

- **Workplace Insights Programme**, UoA Career Development and Employability Services (Aug - Oct 2026, completed): NZ workplace and networking workshops; catch-ups with a University of Auckland alumni volunteer
- **Volunteer**, Auckland Library of Tools (2026 - present); AUSA orientation volunteer
- **Photography Director**, Chinese Postgraduate Society (CNPG), New Zealand (2026 - present)
- **Head of Student News Department**, Huazhong Agricultural University: led news writing and photography, task allocation and deadlines

REFEREES

- Available on request.